

Calculating Expected Scores in R

<https://gemini.google.com/app/47605c4b04729357>

User prompt: I have a data frame in R called df_Elo. It has the columns player_name and diff_EA. I would like to find the rows with the five smallest (most negative) and five largest values for diff_EA and present that information in a chart

Response: Here is the ggplot2 and dplyr solution to filter the top and bottom 5 players and display them in a horizontal bar chart:

```
library(dplyr)
library(ggplot2)

# 1. Get top 5 largest and top 5 smallest diff_EA values
top_bottom_df <- df_Elo %>%
  filter(!is.na(diff_EA)) %>%
  slice_max(order_by = diff_EA, n = 5) %>%
  bind_rows(
    df_Elo %>%
      filter(!is.na(diff_EA)) %>%
      slice_min(order_by = diff_EA, n = 5)
  ) %>%
  distinct() # Ensures no overlap if dataset has fewer than 10 rows

# 2. Create horizontal bar chart
ggplot(top_bottom_df, aes(x = reorder(player_name, diff_EA), y = diff_EA, fill = diff_EA > 0)) +
  geom_col() +
  geom_text(
    aes(label = round(diff_EA, 2)),
    hjust = ifelse(top_bottom_df$diff_EA >= 0, -0.2, 1.2),
    size = 3.5
  ) +
  coord_flip() +
  scale_fill_manual(
    values = c("TRUE" = "#2e7d32", "FALSE" = "#c62828"),
    labels = c("TRUE" = "Overperformed", "FALSE" = "Underperformed"),
    name = "Performance"
  ) +
  labs(
    title = "Top 5 and Bottom 5 Players by diff_EA",
    x = "Player Name",
    y = "Performance Difference (diff_EA)"
  ) +
  theme_minimal() +
  theme(panel.grid.minor = element_blank())
```

Key Elements of the Code:

- **slice_max()** / **slice_min()**: Efficiently extracts the highest 5 and lowest 5 diff_EA values.
- **reorder(player_name, diff_EA)**: Sorts the players on the y-axis by their performance values rather than alphabetical order.
- **coord_flip()**: Flips the bar chart horizontally, making player names much easier to read.
- **geom_text()**: Adds text labels on the bars showing exact values, dynamically positioning them outside positive and negative bars using hjust.

Would you like to customize the colors or add exact rating numbers to the labels as well?

User prompt: Thank you. Could you also show that information in a table?

Response: Here are two clean options for displaying this top 5 and bottom 5 summary table in R—one using standard dplyr output, and one formatted with gt for publication-style visualization.

Option 1: Formatted Table with gt (Recommended)

The gt package creates cleanly formatted tables with custom header grouping and color coding:

```
library(dplyr)
library(gt)

# 1. Filter top 5 and bottom 5, adding a category label
table_df <- df_Elo %>%
  filter(!is.na(diff_EA)) %>%
  mutate(Group = if_else(diff_EA >= 0, "Top 5 (Overperformed)", "Bottom 5 (Underperformed)")) %>%
  group_by(Group) %>%
  slice_max(order_by = abs(diff_EA), n = 5) %>%
  ungroup() %>%
  arrange(desc(diff_EA)) %>%
  select(Group, player_name, diff_EA)

# 2. Render styled table
table_df %>%
  gt(groupname_col = "Group") %>%
  tab_header(
    title = "Player Performance Extremes",
    subtitle = "Top 5 and Bottom 5 players by diff_EA"
  ) %>%
  fmt_number(
    columns = diff_EA,
    decimals = 2
  ) %>%
  cols_label(
    player_name = "Player Name",
```

```

diff_EA = "diff_EA Score"
) %>%
tab_options(
  row_group.font.weight = "bold"
)

```

Option 2: Console Table with dplyr & knitr::kable()

If you need a quick output in the R console or for R Markdown/Quarto reports:

```

library(dplyr)
library(knitr)

# Combine Top 5 and Bottom 5 into a single summary table
table_summary <- df_Elo %>%
  filter(!is.na(diff_EA)) %>%
  arrange(desc(diff_EA)) %>%
  slice(c(1:5, (n() - 4):n())) %>%
  mutate(
    Category = ifelse(row_number() <= 5, "Top 5 (Highest)", "Bottom 5 (Lowest)"),
    diff_EA = round(diff_EA, 3)
  ) %>%
  select(Category, player_name, diff_EA)

# Display formatted Markdown table
kable(table_summary, col.names = c("Category", "Player Name", "diff_EA"))

```

Would you like to add additional player stats (such as total points or pre-rating) to this table summary?

User prompt: I would like to look at a scatter plot of all of the players diff_EA, with the X axis not labeled by name

Response: Here are two ways to plot this in R—one using ggplot2 (with positive/negative color coding and a reference line at 0) and one using Base R.

Option 1: ggplot2 (Recommended)

This hides the player names on the x-axis and uses each player's row index instead, with a dashed reference line at $y = 0$.

```

library(ggplot2)
library(dplyr)

df_Elo %>%
  filter(!is.na(diff_EA)) %>%
  mutate(player_index = row_number()) %>%

```

```

ggplot(aes(x = player_index, y = diff_EA)) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray50") +
  geom_point(aes(color = diff_EA > 0), size = 2.5, alpha = 0.8) +
  scale_color_manual(
    values = c("TRUE" = "#2e7d32", "FALSE" = "#c62828"),
    labels = c("TRUE" = "Overperformed", "FALSE" = "Underperformed"),
    name = "Performance"
  ) +
  labs(
    title = "Scatter Plot of Player diff_EA Scores",
    x = "Players",
    y = "diff_EA Score"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_blank(), # Hides x-axis tick labels (names/numbers)
    axis.ticks.x = element_blank(), # Hides x-axis ticks
    panel.grid.minor = element_blank()
  )

```

Option 2: Sorted Distribution View

If you want to view the scatter plot sorted from lowest to highest diff_EA (creating an S-curve/distribution view), simply order diff_EA before plotting:

```

df_Elo %>%
  filter(!is.na(diff_EA)) %>%
  arrange(diff_EA) %>%
  mutate(rank = row_number()) %>%
  ggplot(aes(x = rank, y = diff_EA, color = diff_EA > 0)) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray50") +
  geom_point(size = 2.5) +
  scale_color_manual(values = c("FALSE" = "#c62828", "TRUE" = "#2e7d32"), guide = "none") +
  labs(
    title = "Sorted Player diff_EA Distribution",
    x = "Rank (Lowest to Highest)",
    y = "diff_EA Score"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_blank(),
    axis.ticks.x = element_blank()
  )

```

Option 3: Base R Quick Plot

```

# Quick base R scatter plot without x-axis labels
clean_diff <- na.omit(df_Elo$diff_EA)

```

```
plot(
  clean_diff,
  pch = 19,
  col = ifelse(clean_diff > 0, "#2e7d32", "#c62828"),
  xaxt = "n", # Suppress x-axis text
  xlab = "Players",
  ylab = "diff_EA Score",
  main = "Scatter Plot of Player diff_EA"
)
abline(h = 0, lty = 2, col = "gray50")
```
